MARK SUBLETTE

Simulation Engineering Student | C++ Developer | AI & Automation

Beaumont, Texas | github.com/marksublette | marksublette.github.io

PROFESSIONAL SUMMARY

Simulation Engineering student at Full Sail University (reported GPA: 4.0) focused on C++, local AI, automation, and software systems. Business owner with project management and technical problem-solving experience. Building orchestration software and researching compression and memory-efficiency techniques.

TECHNICAL SKILLS

Programming: C++, object-oriented programming, HTML, CSS, vectors, references, functions, debugging, input validation; Python (working knowledge).

Tools: Visual Studio, Git, GitHub, Projects, branches, Issues, commits, Pull Requests. Interests: Local AI, simulation, robotics, memory optimization, software architecture.

SELECTED TECHNICAL PROJECTS

SAMANTHA Orchestration Simulator | C++

Developed a console application for best-fit task scheduling across specialized agents using task complexity, capability, specialty, and availability. Includes batch assignment, task lifecycle tracking, and error handling. github.com/marksublette/SAMANTHA-Orchestration-Simulator

SAMANTHA Local AI | Independent Development (ongoing)

Developing a locally operated personal AI system with model integration, task coordination, user-controlled computing, and performance optimization.

NEW EWE | Memory-Efficiency Project (ongoing)

Developing an experimental approach intended to reduce RAM usage and improve memory efficiency. Exploring practical trade-offs among memory consumption, application performance, and responsiveness; results remain under evaluation.

Pressure Memory / Com-Pressure | Independent Research (ongoing)

Researching predictive lossless compression and independently accessible compressed regions, emphasizing exact reconstruction, storage efficiency, and access latency. Manuscript awaiting peer review.

PROFESSIONAL EXPERIENCE

Owner / Operator — Sublette Carpentry | Beaumont, Texas

Managed construction projects, estimating, bids, scheduling, materials, client communication, technical troubleshooting, and project delivery.

EDUCATION

Full Sail University — B.S. Simulation Engineering (in progress) | Reported GPA: 4.0

Relevant study: programming, discrete mathematics, project and portfolio development.

PROFESSIONAL OBJECTIVE

Continue advancing software engineering skills and pursue opportunities in AI, simulation, robotics, and automation.